The Eckmann-Hilton argument: composition on the second loop space is commutative

The Univalent Foundations Program

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Editorial note. The counted statement is the author theorem labelled thm:EckmannHilton (source lines 552-560); the proof span 562-634 is the author's single proof environment, retained line for line.

Editorial note. The proof invokes the right and left unit laws for path concatenation; the retained context carries the author's statement of those laws, so both citations resolve inside the composed document.

Editorial note. The loop-space notation used by the statement is explained in the standard prerequisites; it is ordinary mathematical notation requiring no book-internal citation.

Editorial scope. One original result of the HoTT book chapter cha:basics, counted once: commutativity of composition on the second loop space by the Eckmann-Hilton argument. The counted theorem and its whole author proof are retained; the preceding loop-space commentary and the follow-up remark remain uncounted in the source.

Allowed standard prerequisites (editorial).

- Loop spaces: for a type A with basepoint a, the type of loops x = x, and the second loop space of 2-dimensional loops between identity loops.
- Path concatenation and inversion with their definitional behaviour on refl.
- The unit laws for path concatenation (right and left unit laws as elements of iterated identity types), stated by the retained context.
- The induction principle for identity types (path induction) with its conversion rule.

Lemma 0.0.1. *Suppose* $A : \mathcal{U}$*, that* $x, y, z, w : A$ *and that* $p : x = y$ *and* $q : y = z$ *and* $r : z = w$*. We have the following:*

(i) $p = p \cdot \mathsf{refl}_y$ *and* $p = \mathsf{refl}_x \cdot p$.
(ii) $p^{-1} \cdot p = \mathsf{refl}_y$ *and* $p \cdot p^{-1} = \mathsf{refl}_x$.
(iii) $(p^{-1})^{-1} = p$.
(iv) $p \cdot (q \cdot r) = (p \cdot q) \cdot r$.

Theorem 0.0.2 (Eckmann–Hilton)**.** *The composition operation on the second loop space*

$$\Omega^2(A) \times \Omega^2(A) \to \Omega^2(A)$$

is commutative: $\alpha \cdot \beta = \beta \cdot \alpha$*, for any* $\alpha, \beta : \Omega^2(A)$*.*

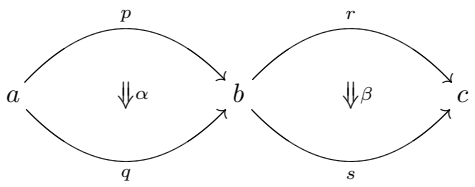
Proof. First, observe that the composition of 1-loops $\Omega A \times \Omega A \to \Omega A$ induces an operation

$$\star : \Omega^2(A) \times \Omega^2(A) \to \Omega^2(A)$$

as follows: consider elements $a, b, c : A$ and 1- and 2-paths,

$$\begin{aligned} p &: a = b, & r &: b = c \\ q &: a = b, & s &: b = c \\ \alpha &: p = q, & \beta &: r = s \end{aligned}$$

as depicted in the following diagram (with paths drawn as arrows).



Composing the upper and lower 1-paths, respectively, we get two paths $p \cdot r,\ q \cdot s : a = c$, and there is then a "horizontal composition"

$$\alpha \star \beta : p \cdot r = q \cdot s$$

between them, defined as follows. First, we define $\alpha \cdot_{\mathsf{r}} r : p \cdot r = q \cdot r$ by path induction on r, so that

$$\alpha \cdot_{\mathsf{r}} \mathsf{refl}_b \equiv {\mathsf{ru}_p}^{-1} \cdot \alpha \cdot \mathsf{ru}_q$$

where $\mathsf{ru}_p : p = p \cdot \mathsf{refl}_b$ is the right unit law from Lemma 0.0.1(i). We could similarly define $\cdot_{\mathsf{r}}$ by induction on α, or on all paths in sight, resulting in different judgmental equalities, but for present purposes the definition by induction on r will make things simpler. Similarly, we define $q \cdot_{\mathsf{l}} \beta : q \cdot r = q \cdot s$ by induction on q, so that

$$\mathsf{refl}_b \cdot_{\mathsf{l}} \beta \equiv {\mathsf{lu}_r}^{-1} \cdot \beta \cdot \mathsf{lu}_s$$

where lu_r denotes the left unit law. The operations $\cdot_{\mathsf{l}}$ and $\cdot_{\mathsf{r}}$ are called **whiskering**. Next, since $\alpha \cdot_{\mathsf{r}} r$ and $q \cdot_{\mathsf{l}} \beta$ are composable 2-paths, we can define the **horizontal composition** by:

$$\alpha \star \beta \ :\equiv\ (\alpha \cdot_{\mathsf{r}} r) \cdot (q \cdot_{\mathsf{l}} \beta).$$

Now suppose that $a \equiv b \equiv c$, so that all the 1-paths p, q, r, and s are elements of $\Omega(A, a)$, and assume moreover that $p \equiv q \equiv r \equiv s \equiv \mathsf{refl}_a$, so that $\alpha : \mathsf{refl}_a = \mathsf{refl}_a$

and $\beta : \mathsf{refl}_a = \mathsf{refl}_a$ are composable in both orders. In that case, we have

$$\begin{aligned}
\alpha \star \beta &\equiv (\alpha \mathbin{\blacktriangleright_{\mathsf{r}}} \mathsf{refl}_a) \blacktriangleright (\mathsf{refl}_a \mathbin{\blacktriangleright_{\mathsf{l}}} \beta) \\
&= {\mathsf{ru}_{\mathsf{refl}_a}}^{-1} \blacktriangleright \alpha \blacktriangleright \mathsf{ru}_{\mathsf{refl}_a} \blacktriangleright {\mathsf{lu}_{\mathsf{refl}_a}}^{-1} \blacktriangleright \beta \blacktriangleright \mathsf{lu}_{\mathsf{refl}_a} \\
&\equiv {\mathsf{refl}_{\mathsf{refl}_a}}^{-1} \blacktriangleright \alpha \blacktriangleright \mathsf{refl}_{\mathsf{refl}_a} \blacktriangleright {\mathsf{refl}_{\mathsf{refl}_a}}^{-1} \blacktriangleright \beta \blacktriangleright \mathsf{refl}_{\mathsf{refl}_a} \\
&= \alpha \blacktriangleright \beta.
\end{aligned}$$

(Recall that $\mathsf{ru}_{\mathsf{refl}_a} \equiv \mathsf{lu}_{\mathsf{refl}_a} \equiv \mathsf{refl}_{\mathsf{refl}_a}$, by the computation rule for path induction.) On the other hand, we can define another horizontal composition analogously by

$$\alpha \star' \beta \;:\equiv\; (p \mathbin{\blacktriangleright_{\mathsf{l}}} \beta) \blacktriangleright (\alpha \mathbin{\blacktriangleright_{\mathsf{r}}} s)$$

and we similarly learn that

$$\alpha \star' \beta = \beta \blacktriangleright \alpha.$$

But, in general, the two ways of defining horizontal composition agree, $\alpha \star \beta = \alpha \star' \beta$, as we can see by induction on α and β and then on the two remaining 1-paths, to reduce everything to reflexivity. Thus we have

$$\alpha \blacktriangleright \beta = \alpha \star \beta = \alpha \star' \beta = \beta \blacktriangleright \alpha. \qquad \square$$